

4G - Who is paying your cellular phone

Silke Hottnann
Isha Singh

Nokia Bell Labs

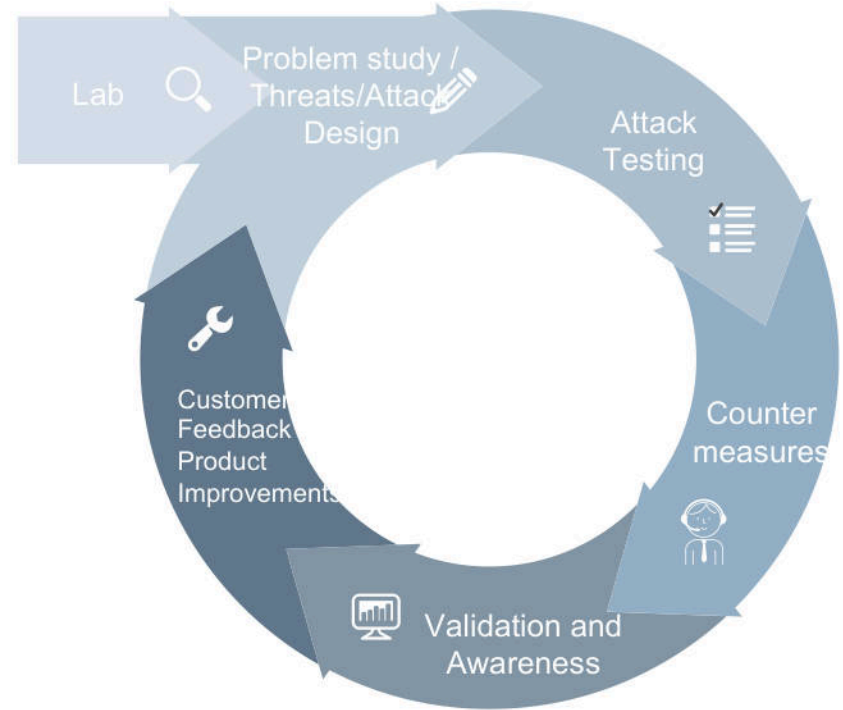
Industrial Security Research?

Nokia Bell Labs – Future Attacks and Mitigation

Research that solves real problems together with our customers and sometimes even competitors

- Theoretical studies go into attack and countermeasure design
- Validation and awareness of our research by GSMA standards input and publication
- Customer feedback and test results allow us to fine-tune and optimize our countermeasures
- Research input will fit product needs and operators requests
- Operator needs can be discovered "live" for new research challenges and disruptive new solutions

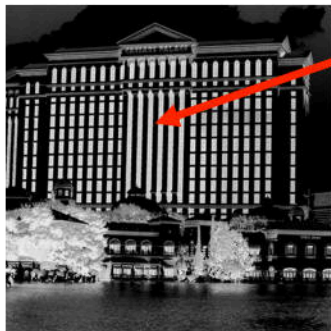
Bell Labs Research Lifecycle



You connect
What does actually
happen?

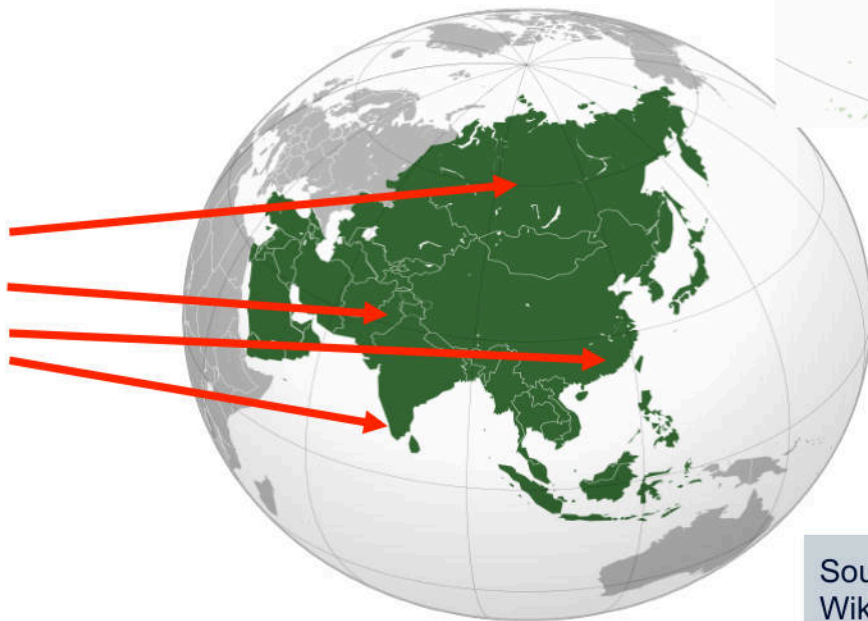
Roaming

Why should you care?



You connected to AT&T,
Verizon, T-Mobile, Sprint

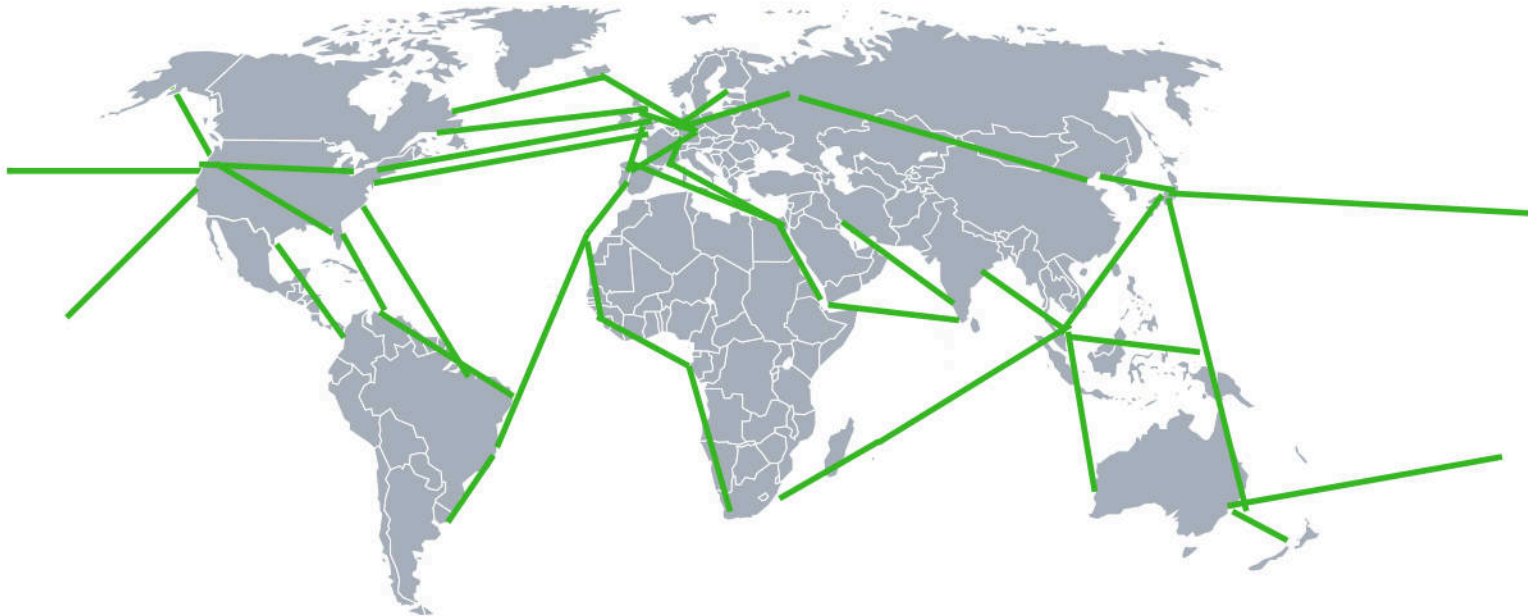
DefCon participants
CMCC, Airtel,
MegaFon, Telenor



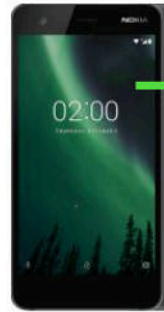
My colleagues,
friends, family
connected to DNA,
Elisa, Telia

Connecting networks – The hidden private Internet

The Interconnection Network (IPX)



I switch on my phone



Las Vegas
Antenna



Core Network



Carrier / IPX

Authentication -> run to home network
Checking subscriber

Carrier / IPX



Core Network



Bell Labs

What is this secret network?
Where does it come from?

The good auld history

1981 – Nordic Operator Meeting

Need to call my wife, she has birthday today. Stupid that I can not use my mobile.

The sauna is not hot enough. Lets connect the networks. So you call me and I can heat up the sauna before you arrive.

The beer is warm.... Should be longer in the fridge.



Starting of practical discussions

We would need a cable under the water of the baltic sea...

Cold beer would be nicer. Yeah we should connect the networks.....then you can put it before in the fridge in time.



The technical part worked out

The networks are owned by the governments anyway.

We don't need security. It is a closed network just for us

I know someone in ITU who can help

We could invite some other operators.

People will love it. Pizza delivery everywhere.

Some new protocol is needed for this.

Can you pass me another beer and the mustard?



Evolutions of IPX

- Started with 5 Nordic operators and calls only about 35 years ago
- Now about 2000 companies connected to it
 - Mobile operators
 - Service providers (SMS aggregators, password recovery)
 - Satellite communication providers etc
- Very inhomogenous operator structure
- Networks are a mix and match
 - 2G, 2.5G, 3G, 4G and now 5G
 - Different hardware, protocols, products, releases
 - Many services voice, SMS, MMS, IMS, data, VoIP
- Network evolved, but security awareness only recently started (2014)



Help & Customer Service

Reset Your Password

If you've forgotten your password, you can reset it by requesting that a personalized "password reset" link be sent to your e-mail address.



To make this update from a computer that you haven't used previously to shop on our website, we'll also ask you to confirm all of your account information before being able to create a new password.

To request a password reset:

1. Go to [Forgot Your Password](#).
 2. Follow the on-screen instructions.
 3. Follow the instructions in the e-mail sent to the e-mail address on your account.
- Note:** If you created your account with a mobile phone number, you'll receive an SMS message with instructions instead of an e-mail.

SMS providers

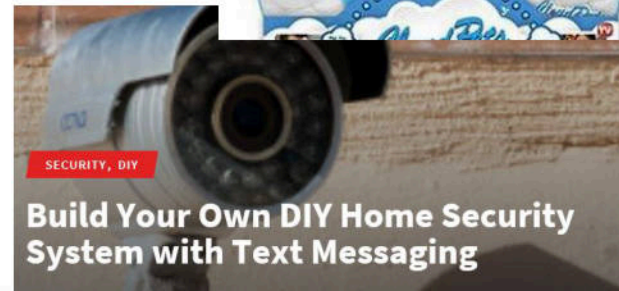
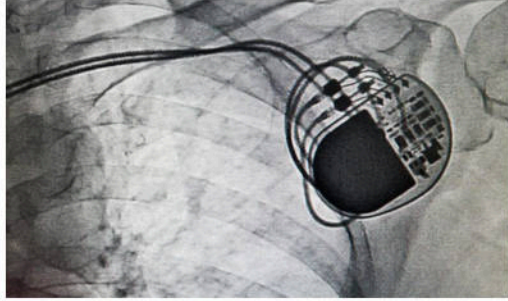
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SEP – Somebody Else Problem?

Message for you.....



It is not only you that is "reachable"



Build Your Own DIY Home Security System with Text Messaging

Security?

Who would hack this network

KIM ZETTER SECURITY 04.28.16 07:00 AM

THE CRITICAL HOLE AT THE HEART OF OUR CELL PHONE NETWORKS



How the NSA could spy on a American phone — without congressional approval



TRACK ANY MOBILE
a service for private investigators

Logged in as:
Email: mail@trackanymobile.com
Mobile: 447941000000
[Logout](#)

TRACK ANYONE WITH A GSM MOBILE PHONE TO AN ACCURACY OF ~3 MILES

✓ Access to phone not required ✓ Additional hardware not required

HOME USER FAQ CONTACT

User Home Track Phones Manage Localities Settings Purchase Credits

Currently defined localities (a locality is a geographic area generally covering a few miles).

Name	Network Operator	Country	Added	
Bond Street / Oxford Circus Area	T-Mobile [234 3...	United Kingdom	2008-03-04 10:16:27	Add Locality
Eltham / Mottingham Area	T...			Edit Delete
London Bridge Station Area	T...			
Depford Area	T...			

[Authentication](#), [Fraud](#), [Phishing](#)

Bank Account Hackers Used SS7 to Intercept Security Codes

Well-Known Signaling System 7 Protocol Flaws Exploited in Germany

Verified SMS/OTP

03 / VA Service

communications with partners and employees are exposed to interception. When SMS and factor authentication (2FA) are used for recovery, it can also be leveraged by criminals to commit fraud

Too often organizations are unaware of the exposure of current business processes to Cellular (SS7 and MITM) interception. Vaulto offers the world's first telecom vulnerability assessment (VA) service, to

SkyLock's location finding capabilities are based on the ability to send and handle standard signaling messages (MAP messages) through the international SS7 network. This solution does not require any

NOKIA

Existing Attacks for the "old" SS7

Most of the attacks today are still SS7 – but things change

- Location Tracking
- Eavesdropping
- Fraud
- Denial of Service user & network
- Credential theft
- Data session hijacking
- Unblock
- SMS interception
- One time password account takeover for WhatsApp, Facebook, Whatsapp, bitcoin wallet

Hackers Exploit SS7 Flaws to Loot Bank Accounts

Kovacs

Request: 1 Cell - GSM

```
{
  "token": "101472903251",
  "radius": "00",
  "msc": "00",
  "cellid": "1",
  "lmsi": "1",
  "address": "1"
}
```

Response:

```
{
  "status": "ok",
  "radius": "00",
  "msc": "00",
  "cellid": "1",
  "lmsi": "1",
  "address": "1"
}
```



Iranian company - Global Operator in Wiretapping

It has listed itself as a virtual operator to intercept keys from the SS7 inter-operator network.

December 19, 2015

The Little Black Book of Billionaire

Stunt With

here has been shaken
od to intercept 3G

net traffic in the 3G
ossible without the
or encrypting the

The used to date popular interception method for breaking in and eavesdropping on

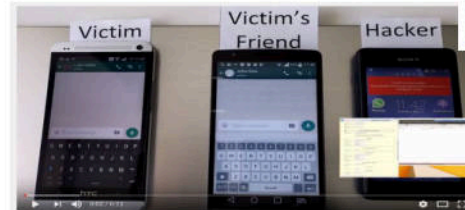
WIRELESS

Telenor mobile network hit by international signal

Monday 22 February 2016 | 16:03 CET | News

Telenor said it suffered a major mobile network outage for several hours due to incorrect signalling data from an international operator. Service

Media: officials fired for using WhatsApp, Viber and Telegram



WhatsApp SS7 attack

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How do attackers get in

Rent a Service

Become an Operator

Kick in the door



Bribing and Employed

Hack via Internet

Social Engineering

That is how they get in

Well, of course there might be legitimate reason...maybe....

59.0

View Raw Data

Internet Scanner

City	
Country	Some big Asian country
Organization	
ISP	
Last Update	2018-06-27T11:32:02.803916
ASN	AS134

30793

udp

gtp-v1

GPRS Tunneling Protocol

Version: 1

GPRS Tunneling Protocol

Correct data length for version 1

Version: 1

Flags: XXX1 0010

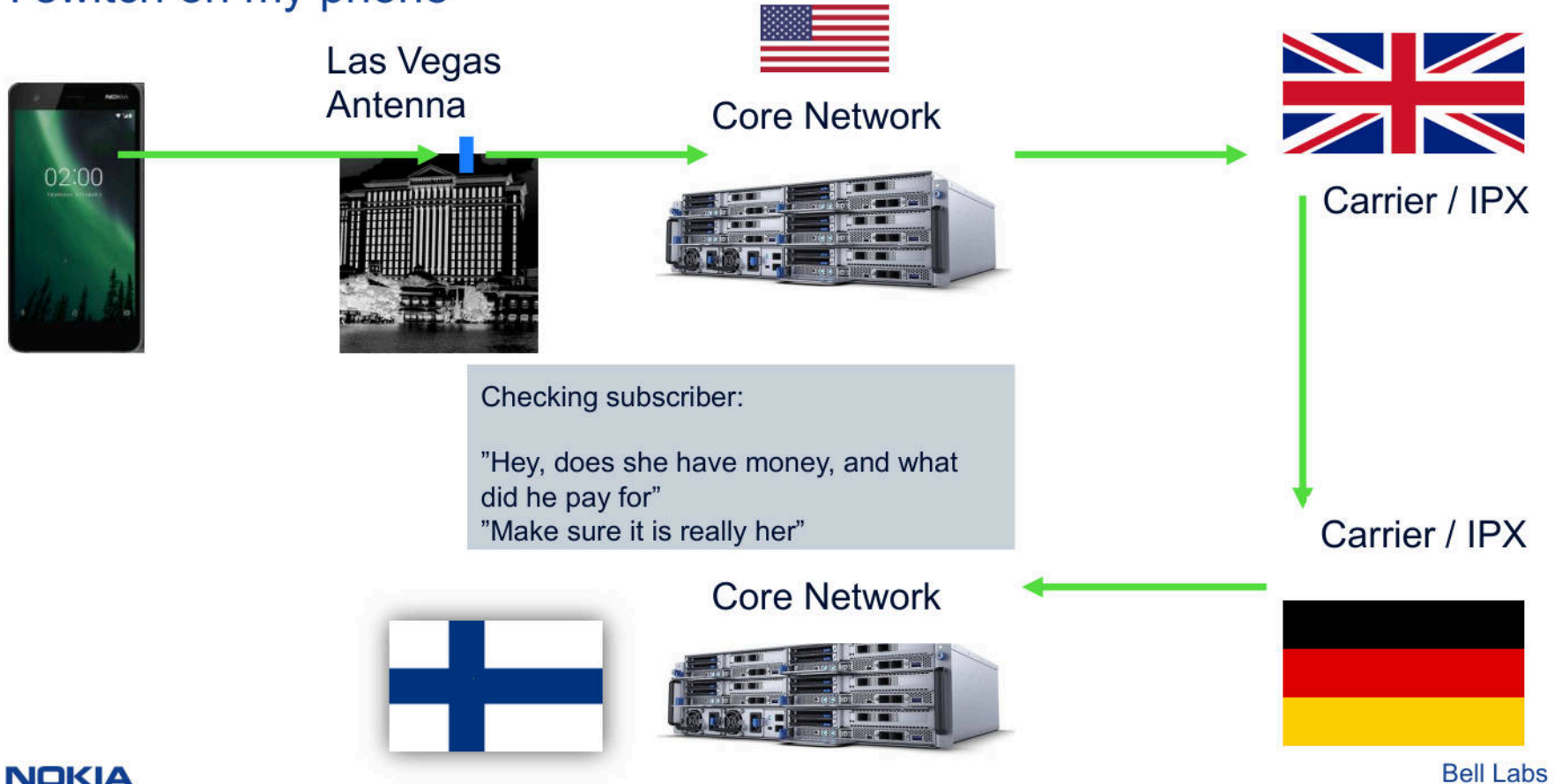
Type: 2 (Echo response)

Length: 6

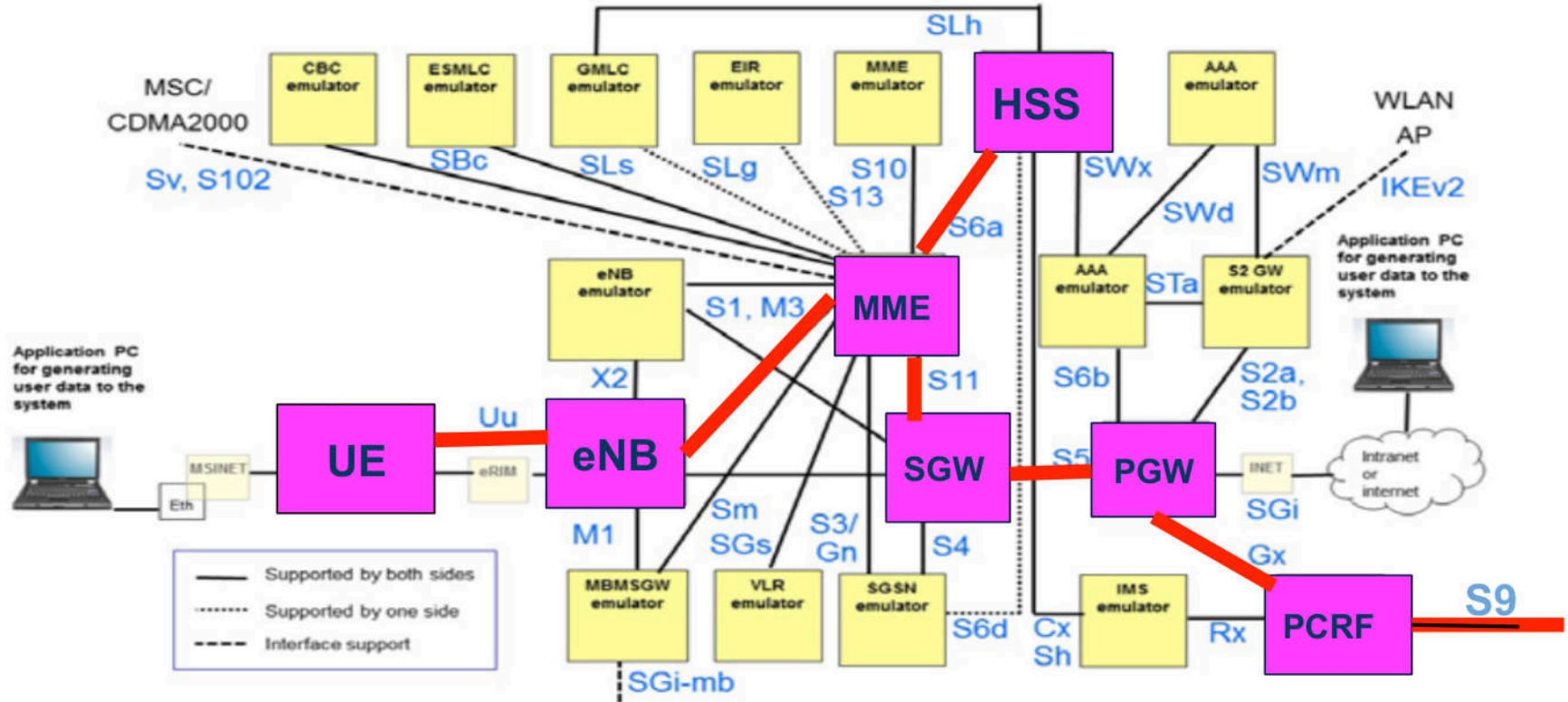
Data: \x0c=\x00\x00\x0e\x00

New protocol - New luck?

I switch on my phone

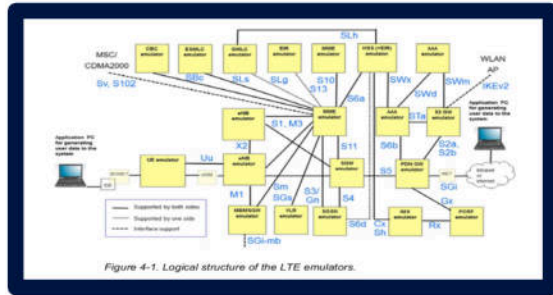


Network used for testing of attack

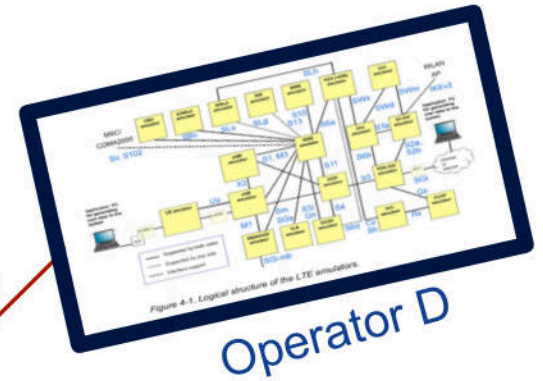


LTE Emulator

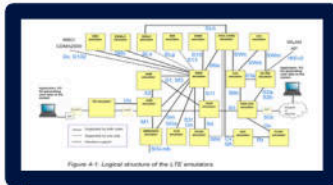
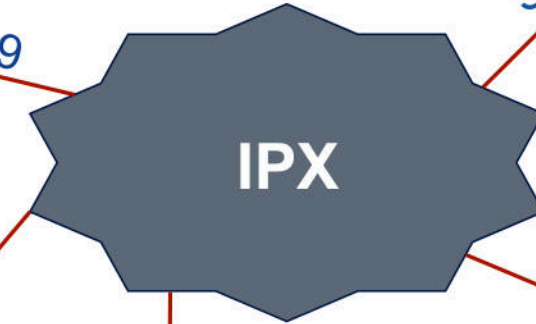
Operators with connected S9 billing interface



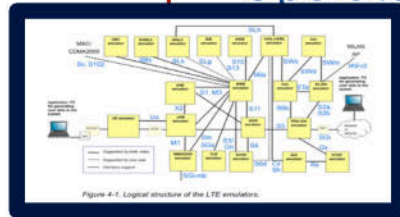
Operator A



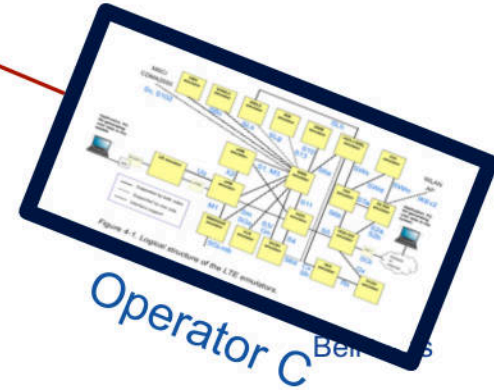
Operator D



Operator B

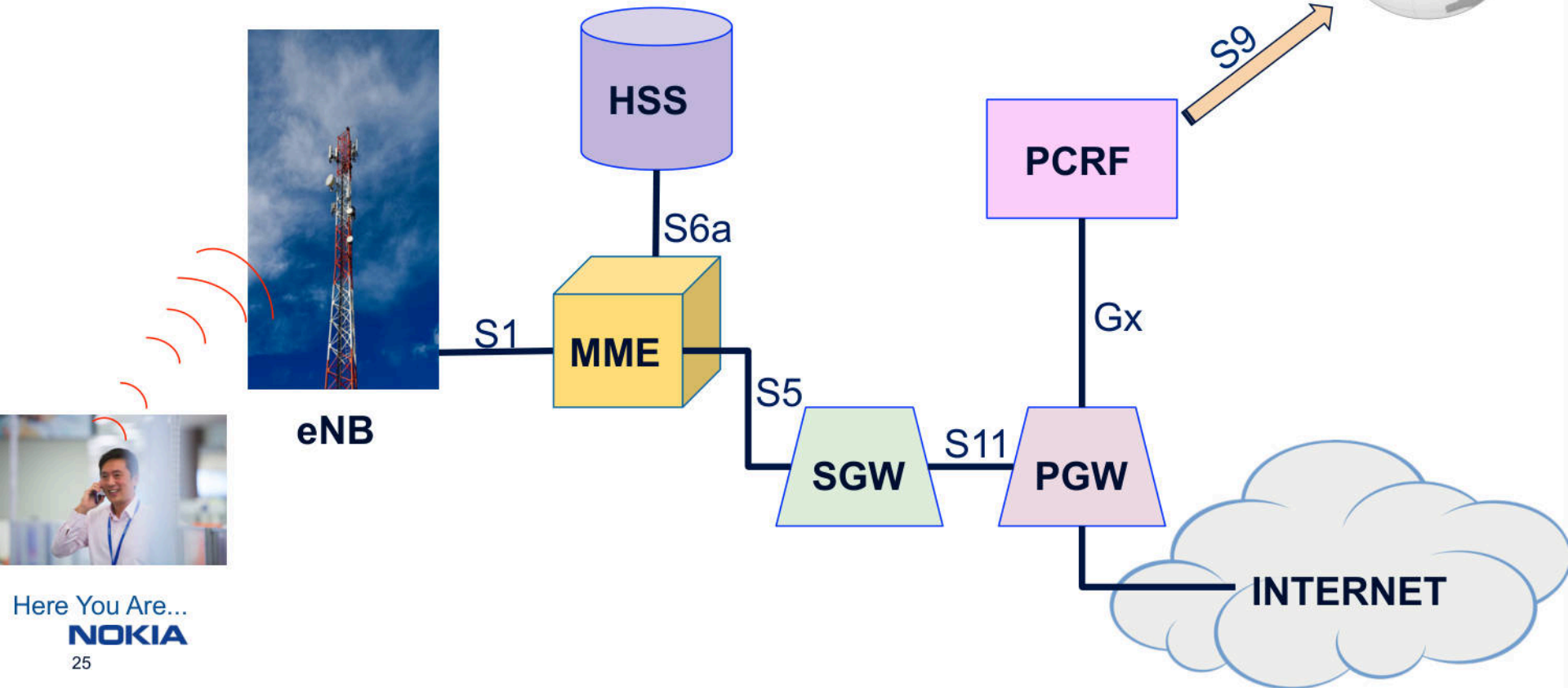


Operator F



Operator C

Network Architecture (3G/4G)



UEPROC [1] ./ueproc 1

```
00111
29.6.2018 11:35:27,512,393 UE-C-1 RRC ENB-C-1 MESSAGE SENT RRC_UL_INFORMATION_T
ANSFER
Waiting for inputs ...
29.6.2018 11:35:27,629,061 UE-C-1 RRC ENB-C-1 MESSAGE RECEIVED RRC_UL_INFORMATIO
N_TRANSFER
29.6.2018 11:35:27,629,818 UE-C-1 NAS PROCEDURE COMPLETED DETACH IMSI=5887110020
00111
29.6.2018 11:35:27,836,066 UE-C-1 RRC ENB-C-1 MESSAGE RECEIVED RRC_CONNECTION_RE
LEASE
```

UEUPROC [1] ./ueuproc 1

```
: .. Startup macro for the system. Not used by the user.

: set echo off
29.6.2018 11:21:25,211,737 UE-U-1 - PROCESS STARTED
Waiting for inputs ...
29.6.2018 11:33:20,722,736 UE-U-1 SAI PROCEDURE COMPLETED L1_CELL_SEARCH "ENB-ID
= 1"
```

SGW [1] ./sgw 1

```
29.6.2018 11:35:27,624,225 SGW-1 GTP_S5 PROCEDURE COMPLETED DELETE_SESSION IMSI=
588711002000111
29.6.2018 11:35:27,624,257 SGW-1 GTP_S11 PROCEDURE COMPLETED DELETE_SESSION IMSI
=588711002000111 R-TEID=4338 S-TEID=4343
29.6.2018 11:35:27,624,318 SGW-1 GTP_S11 MME-1 MESSAGE SENT GTPV2_PDU_DELETE_SES
SION_RESPONSE
```

ERIM [1] ./erim 1

```
: set echo off
29.6.2018 11:21:20,276,273 ERIM-1 - PROCESS STARTED

Running in SAI mode
Waiting for inputs ...
29.6.2018 11:21:24,932,897 ERIM-1 SAI ENB-U-1 MESSAGE RECEIVED ENB_REGISTER
```

PGW [1] ./pgw 1

```
SI=588711002000111
29.6.2018 11:35:27,623,186 PGW-1 DIAMETER_Gx PCRF-1 MESSAGE SENT DIAMETER_GX_CCR
IMSI=588711002000111
29.6.2018 11:35:27,623,485 PGW-1 GTP_S5 PROCEDURE COMPLETED DELETE_SESSION IMSI=
588711002000111
29.6.2018 11:35:27,623,628 PGW-1 GTP_S5 SGW-1 MESSAGE SENT GTPV2_PDU_DELETE_SESS
ION_RESPONSE
29.6.2018 11:35:27,629,852 PGW-1 DIAMETER_Gx PCRF-1 MESSAGE RECEIVED DIAMETER_GX
_CCA IMSI=588711002000111
29.6.2018 11:35:27,629,974 PGW-1 DIAMETER_Gx PROCEDURE COMPLETED CREDIT_CONTROL
IMSI=588711002000111
```

ENBC [1] ./enbc 1

```
29.6.2018 11:35:27,627,868 ENB-C-1 RRC UE-C-1 MESSAGE SENT RRC_UL_INFORMATION_T
ANSFER
29.6.2018 11:35:27,828,162 ENB-C-1 S1-AP MME-1 MESSAGE RECEIVED UE_CONTEXT_RELEA
SE_COMMAND
29.6.2018 11:35:27,828,257 ENB-C-1 S1-AP PROCEDURE STARTED UE_CONTEXT_RELEASE
29.6.2018 11:35:27,829,358 ENB-C-1 RRC UE-C-1 MESSAGE SENT RRC_CONNECTION_RELEAS
E
29.6.2018 11:35:27,830,145 ENB-C-1 S1-AP PROCEDURE COMPLETED UE_CONTEXT_RELEASE
29.6.2018 11:35:27,830,235 ENB-C-1 S1-AP MME-1 MESSAGE SENT UE_CONTEXT_RELEASE_C
OMplete
```

ENBU [1] ./enbu 1

```
: set echo off
29.6.2018 11:21:23,973,567 ENB-U-1 - PROCESS STARTED
Waiting for inputs ...
29.6.2018 11:21:24,992,414 ENB-U-1 SAI ERIM-1 MESSAGE SENT ENB_REGISTER
29.6.2018 11:33:20,722,856 ENB-U-1 SAI PROCEDURE COMPLETED L1_CELL_SEARCH "UE-ID
= 1"
```

MME [1] ./mme 1

```
29.6.2018 11:35:27,826,299 MME-1 S1-AP PROCEDURE STARTED UE_CONTEXT_RELEASE
29.6.2018 11:35:27,826,340 MME-1 S1-AP ENB-C-1 MESSAGE SENT UE_CONTEXT_RELEASE_C
OMMAND
29.6.2018 11:35:27,832,481 MME-1 S1-AP ENB-C-1 MESSAGE RECEIVED UE_CONTEXT_RELEA
SE_COMPLETE
29.6.2018 11:35:27,832,618 MME-1 S1-AP PROCEDURE COMPLETED UE_CONTEXT_RELEASE
```

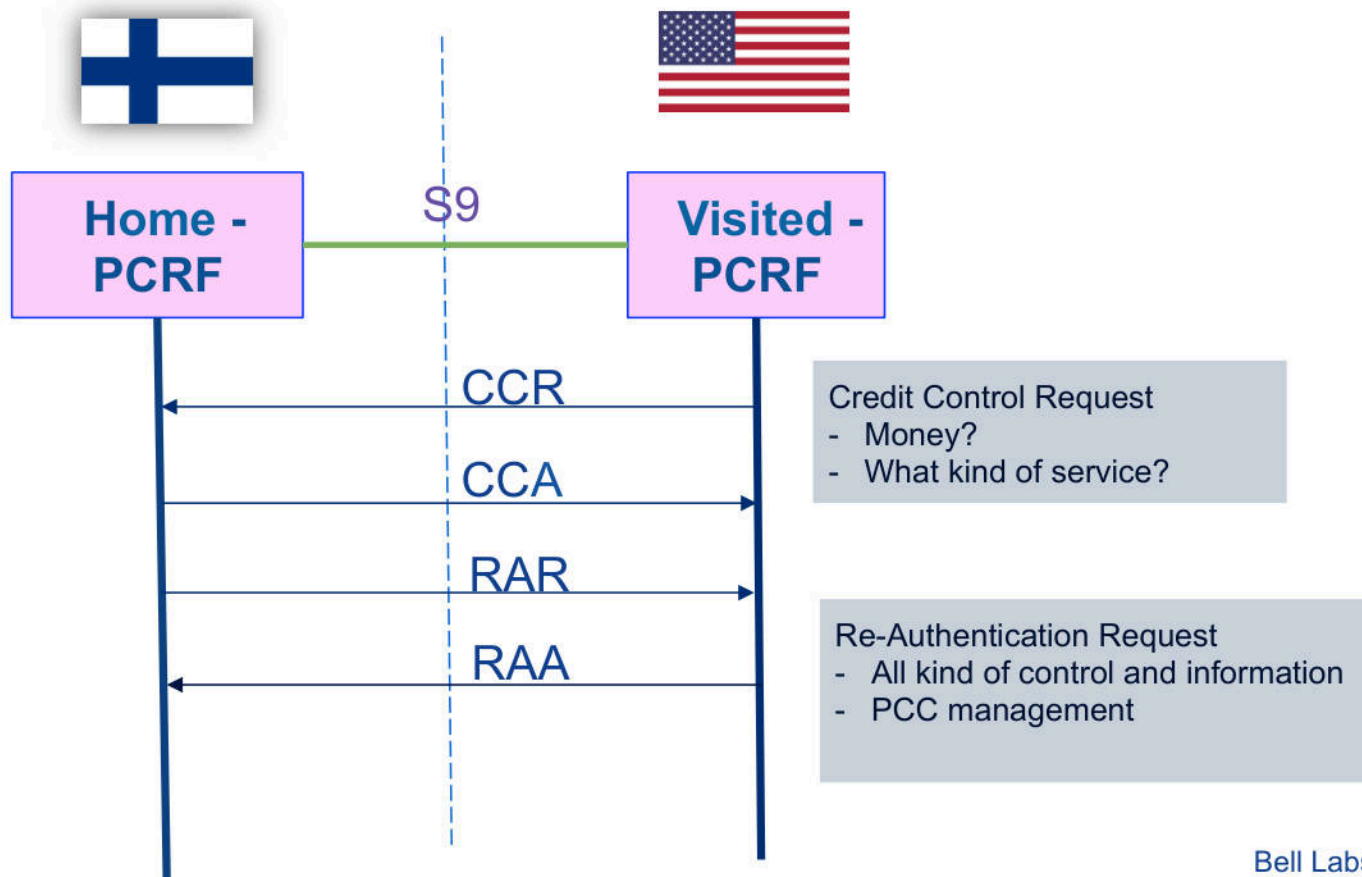
HSS [1] ./hss 1

```
29.6.2018 11:33:49,729,781 HSS-1 DIAMETER MME-1 MESSAGE SENT DIAMETER_DWA
29.6.2018 11:34:19,732,896 HSS-1 DIAMETER MME-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 11:34:19,733,163 HSS-1 DIAMETER MME-1 MESSAGE SENT DIAMETER_DWA
29.6.2018 11:34:49,736,283 HSS-1 DIAMETER MME-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 11:34:49,736,471 HSS-1 DIAMETER MME-1 MESSAGE SENT DIAMETER_DWA
29.6.2018 11:35:19,738,616 HSS-1 DIAMETER MME-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 11:35:19,739,042 HSS-1 DIAMETER MME-1 MESSAGE SENT DIAMETER_DWA
```

PCRF [1] ./pcrf 1

```
29.6.2018 11:34:50,703,817 PCRF-1 DIAMETER PGW-1 MESSAGE RECEIVED DIAMETER_DWA
29.6.2018 11:35:20,706,872 PCRF-1 DIAMETER PGW-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 11:35:20,707,151 PCRF-1 DIAMETER PGW-1 MESSAGE SENT DIAMETER_DWA
29.6.2018 11:35:27,625,510 PCRF-1 DIAMETER_Gx PGW-1 MESSAGE RECEIVED DIAMETER_GX
_CCR IMSI=588711002000111
29.6.2018 11:35:27,625,583 PCRF-1 DIAMETER_Gx PROCEDURE STARTED CREDIT_CONTROL I
MSI=588711002000111
29.6.2018 11:35:27,627,632 PCRF-1 DIAMETER_Gx PROCEDURE COMPLETED CREDIT_CONTROL
IMSI=588711002000111
29.6.2018 11:35:27,627,849 PCRF-1 DIAMETER_Gx PGW-1 MESSAGE SENT DIAMETER_GX_CCA
IMSI=588711002000111
```

Normal incoming request for roaming (Fin in US)

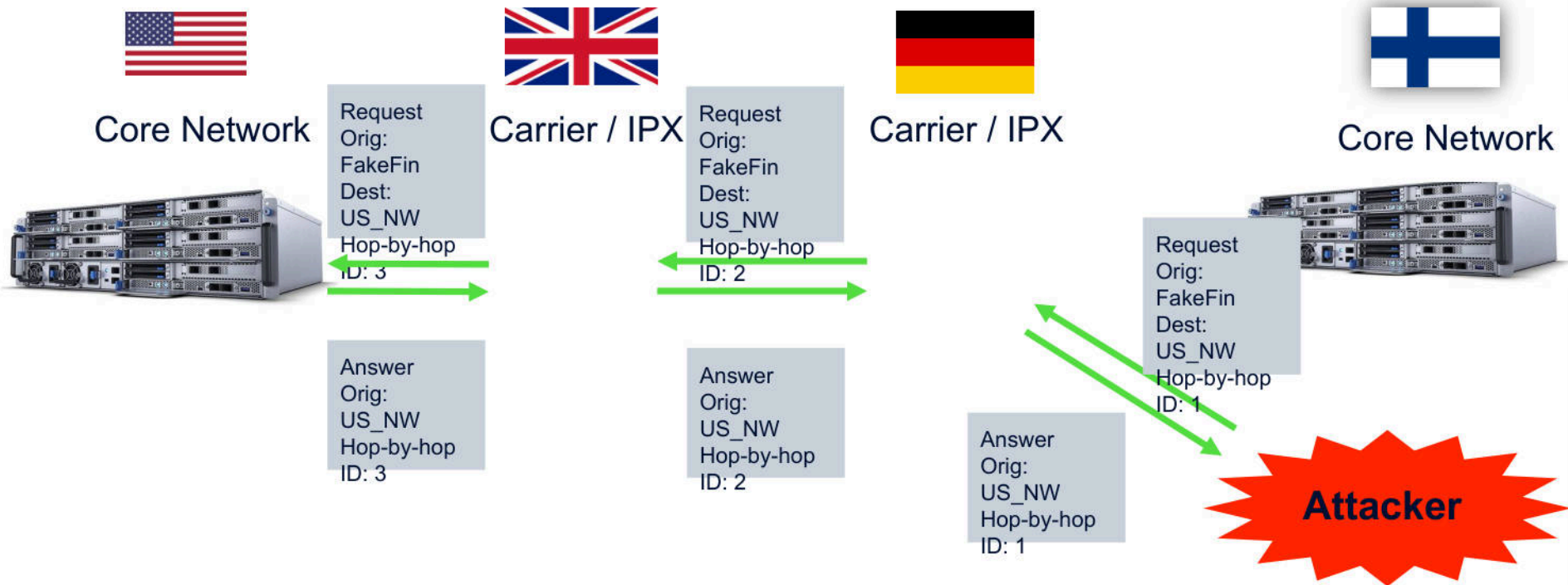


What is a "PCC"?

Something you all have

- Policy Charging Control
 - Defines everything about your subscription
 - Data type
 - Data rates
 - Whatever cellular service you can think off
- Defines how to handle you and what to grant you "service flow filters"
- Usually identified by a string
- My own subscription is company paid and quite "generous"
 - Perfect target for an attacker

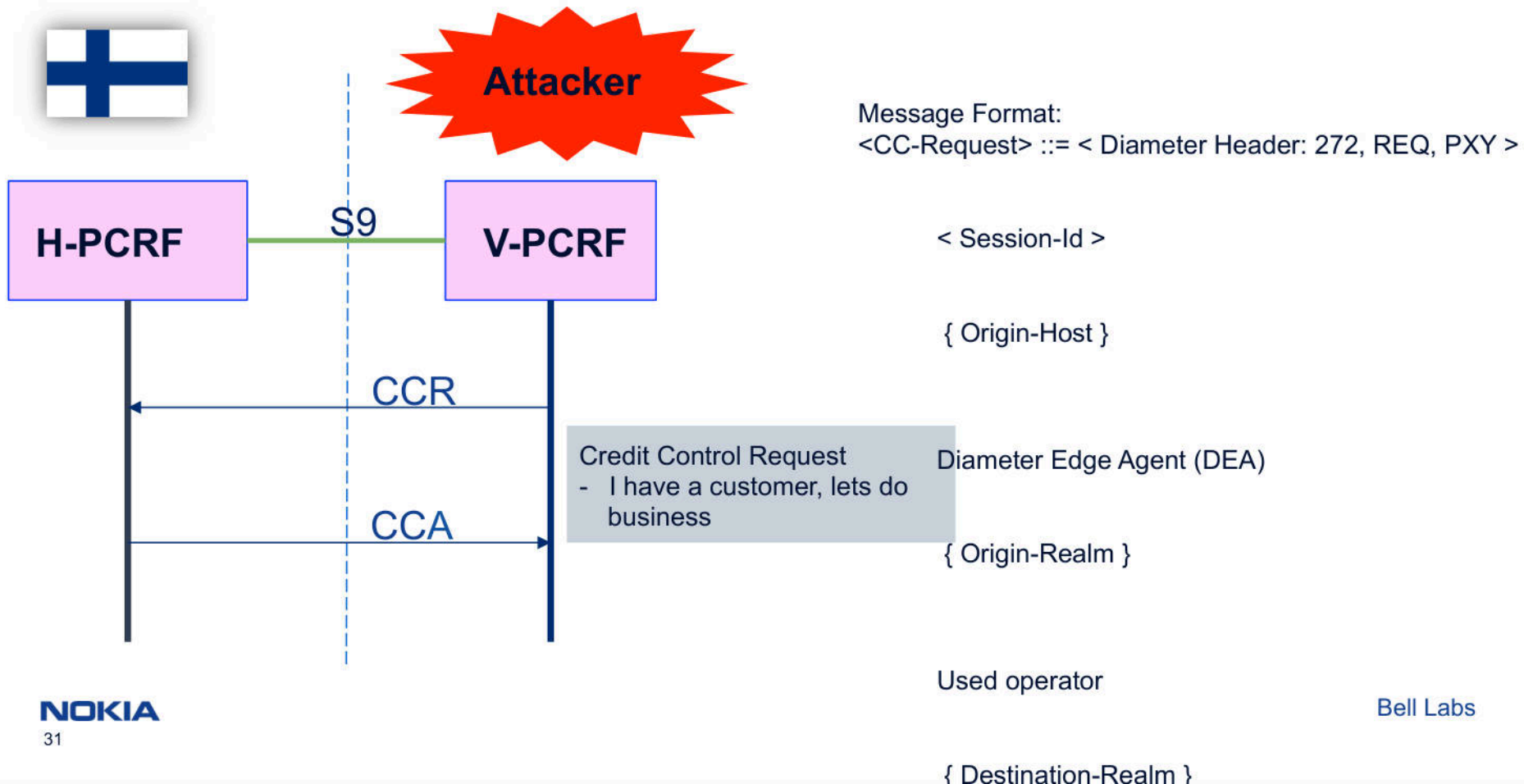
Diameter Routing Issue – Two Possibilities how to route.....Hop-by-Hop



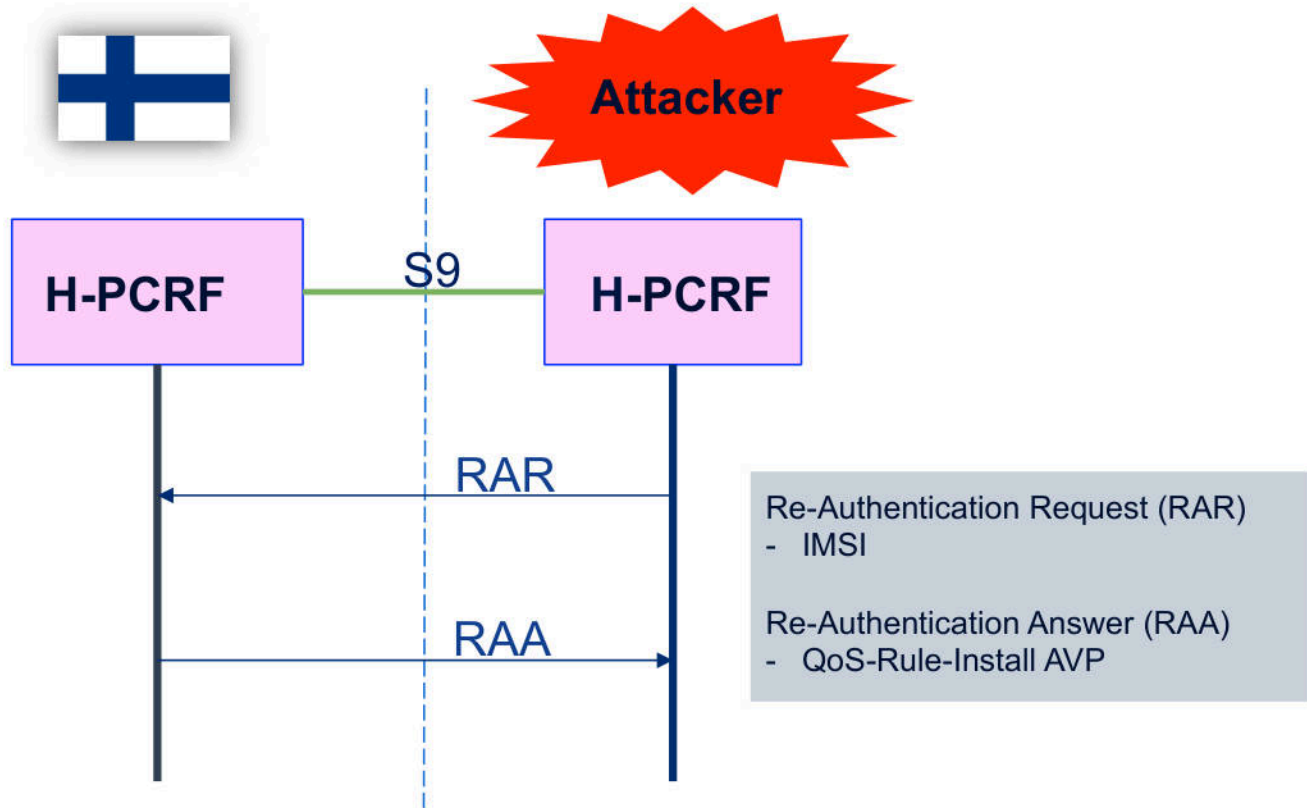
Attack

1. Steal PCC of good subscription
2. Update cheap subscription with PCC of good subscription

Attack scenario against finnish operator – Request PCC via CCR



Requesting PCC via RAR (posing as home network)



UEPROC [1] ./ueproc 1

```
29.6.2018 11:36:05.015.233 UE-C-1 NHS PROCEDURE COMPLETED ATTACH IMSI=588711002000111
29.6.2018 11:36:05.016.253 UE-C-1 RRC ENB-C-1 MESSAGE SENT RRC_UL_INFORMATION_TRANSFER
29.6.2018 11:36:05.017.142 UE-C-1 RRC ENB-C-1 MESSAGE RECEIVED RRC_UL_CAPABILITY_ENQUIRY
29.6.2018 11:36:05.017.258 UE-C-1 RRC ENB-C-1 MESSAGE SENT RRC_UL_CAPABILITY_INFORMATION
29.6.2018 11:36:05.265.910 UE-C-1 RRC ENB-C-1 MESSAGE RECEIVED RRC_DL_INFORMATION_TRANSFER
```

UEUPROC [1] ./ueuproc 1

```
: set echo off
29.6.2018 11:21:25.211.737 UE-U-1 - PROCESS STARTED
Waiting for inputs ...
29.6.2018 11:33:20.722.736 UE-U-1 SAI PROCEDURE COMPLETED L1_CELL_SEARCH "ENB-ID = 1"
29.6.2018 11:36:03.906.972 UE-U-1 SAI PROCEDURE COMPLETED L1_CELL_SEARCH "ENB-ID = 1"
```

SGW [1] ./sgw 1

```
29.6.2018 11:36:05.261.957 SGW-1 GTP_S11 PROCEDURE STARTED MODIFY_BEARER IMSI=588711002000111 R-TEID=8439 S-TEID=8434
29.6.2018 11:36:05.262.010 SGW-1 GTP_S11 PROCEDURE COMPLETED MODIFY_BEARER IMSI=588711002000111 R-TEID=8434 S-TEID=8439
29.6.2018 11:36:05.262.069 SGW-1 GTP_S11 MME-1 MESSAGE SENT GTPV2_PDU_MODIFY_BEARER_RESPONSE
```

ERIM [1] ./erim 1

```
: set echo off
29.6.2018 11:21:20.276.273 ERIM-1 - PROCESS STARTED

Running in SAI mode
Waiting for inputs ...
29.6.2018 11:21:24.992.897 ERIM-1 SAI ENB-U-1 MESSAGE RECEIVED ENB_REGISTER
```

PGW [1] ./pgw 1

```
29.6.2018 12:48:53.489.872 PGW-1 DIAMETER PCRF-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:48:53.490.680 PGW-1 DIAMETER PCRF-1 MESSAGE SENT DIAMETER_DWR
29.6.2018 12:48:53.495.542 PGW-1 DIAMETER PCRF-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:49:23.496.951 PGW-1 DIAMETER PCRF-1 MESSAGE SENT DIAMETER_DWR
29.6.2018 12:49:23.498.774 PGW-1 DIAMETER PCRF-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:49:53.528.635 PGW-1 DIAMETER PCRF-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:49:53.528.890 PGW-1 DIAMETER PCRF-1 MESSAGE SENT DIAMETER_DWR
29.6.2018 12:50:23.557.715 PGW-1 DIAMETER PCRF-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:50:23.558.033 PGW-1 DIAMETER PCRF-1 MESSAGE SENT DIAMETER_DWR
29.6.2018 12:50:53.582.736 PGW-1 DIAMETER PCRF-1 MESSAGE SENT DIAMETER_DWR
29.6.2018 12:50:53.584.657 PGW-1 DIAMETER PCRF-1 MESSAGE RECEIVED DIAMETER_DWR
```

ENBPROC [1] ./enbproc 1

```
_INFORMATION
29.6.2018 11:36:05.057.160 ENB-C-1 S1-AP PROCEDURE COMPLETED INITIAL_CONTEXT_SETUP
29.6.2018 11:36:05.057.368 ENB-C-1 S1-AP MME-1 MESSAGE SENT INITIAL_CONTEXT_SETUP_RESPONSE
29.6.2018 11:36:05.057.886 ENB-C-1 S1-AP MME-1 MESSAGE SENT UPLINK_NAS_TRANSPORT
29.6.2018 11:36:05.263.828 ENB-C-1 S1-AP MME-1 MESSAGE RECEIVED DOWNLINK_NAS_TRANSPORT
29.6.2018 11:36:05.264.621 ENB-C-1 RRC UE-C-1 MESSAGE SENT RRC_DL_INFORMATION_TRANSFER
```

ENBU [1] ./enbu 1

```
29.6.2018 11:21:23.973.567 ENB-U-1 - PROCESS STARTED
Waiting for inputs ...
29.6.2018 11:21:24.992.414 ENB-U-1 SAI ERIM-1 MESSAGE SENT ENB_REGISTER
29.6.2018 11:33:20.722.856 ENB-U-1 SAI PROCEDURE COMPLETED L1_CELL_SEARCH "UE-ID = 1"
29.6.2018 11:36:03.907.125 ENB-U-1 SAI PROCEDURE COMPLETED L1_CELL_SEARCH "UE-ID = 1"
```

MME [1] ./mme 1

```
29.6.2018 12:49:50.449.263 MME-1 DIAMETER HSS-1 MESSAGE SENT DIAMETER_DWR
29.6.2018 12:49:50.450.267 MME-1 DIAMETER HSS-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:50:20.454.139 MME-1 DIAMETER HSS-1 MESSAGE SENT DIAMETER_DWR
29.6.2018 12:50:20.456.874 MME-1 DIAMETER HSS-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:50:50.465.017 MME-1 DIAMETER HSS-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:50:50.465.230 MME-1 DIAMETER HSS-1 MESSAGE SENT DIAMETER_DWR
```

HSS [1] ./hss 1

```
29.6.2018 12:49:50.449.481 HSS-1 DIAMETER MME-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:49:50.449.705 HSS-1 DIAMETER MME-1 MESSAGE SENT DIAMETER_DWR
29.6.2018 12:49:50.450.675 HSS-1 DIAMETER MME-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:50:20.455.383 HSS-1 DIAMETER MME-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:50:20.455.680 HSS-1 DIAMETER MME-1 MESSAGE SENT DIAMETER_DWR
29.6.2018 12:50:50.464.362 HSS-1 DIAMETER MME-1 MESSAGE SENT DIAMETER_DWR
29.6.2018 12:50:50.465.891 HSS-1 DIAMETER MME-1 MESSAGE RECEIVED DIAMETER_DWR
```

PCRF [1] ./pcrf 1

```
29.6.2018 12:48:53.490.544 PCRF-1 DIAMETER PGW-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:48:53.492.359 PCRF-1 DIAMETER PGW-1 MESSAGE SENT DIAMETER_DWR
29.6.2018 12:48:53.495.198 PCRF-1 DIAMETER PGW-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:49:23.497.692 PCRF-1 DIAMETER PGW-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:49:23.497.891 PCRF-1 DIAMETER PGW-1 MESSAGE SENT DIAMETER_DWR
29.6.2018 12:49:53.527.753 PCRF-1 DIAMETER PGW-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:49:53.529.710 PCRF-1 DIAMETER PGW-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:50:23.556.427 PCRF-1 DIAMETER PGW-1 MESSAGE SENT DIAMETER_DWR
29.6.2018 12:50:23.558.943 PCRF-1 DIAMETER PGW-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:50:53.583.529 PCRF-1 DIAMETER PGW-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 12:50:53.583.729 PCRF-1 DIAMETER PGW-1 MESSAGE SENT DIAMETER_DWR
```


Wireshark interface showing a packet capture of Diameter protocol messages. The filter is set to 'diameter'. The packet list shows several messages, with packet 679 selected. The packet details pane shows the structure of the selected packet.

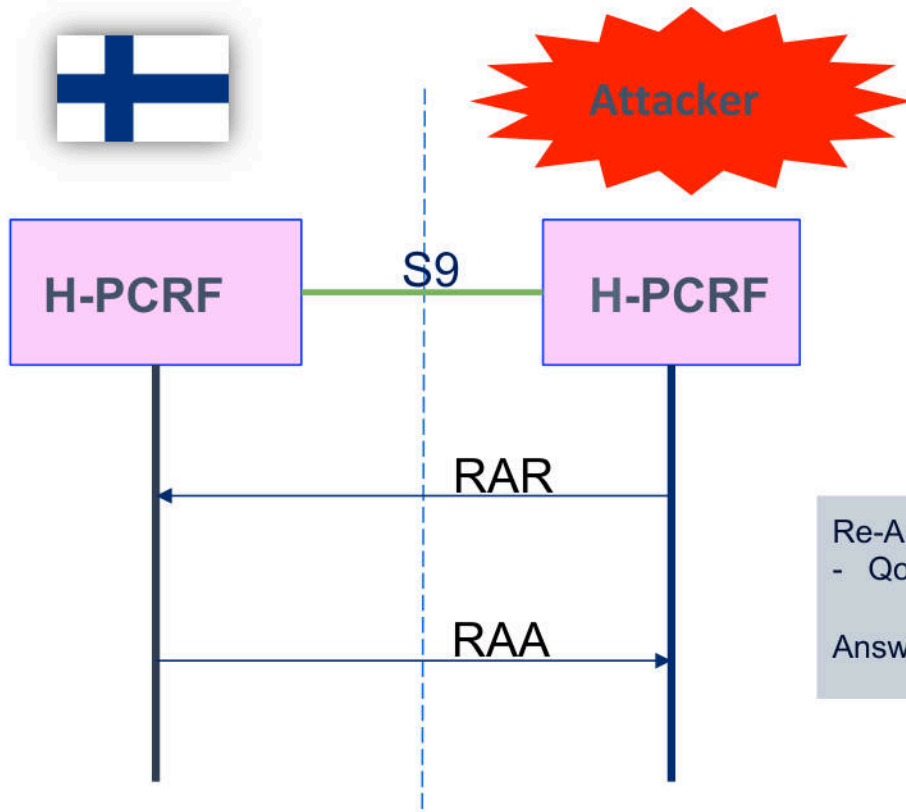
No.	Time	Source	Destination	Protocol	Length	Info
365	47.5750	127.0.0.1	127.0.0.1	DIAMETER	140	cmd-Device-Watchdog Request(280) flags=R--- appl=Diameter Common Messages(0) h2h=7660fed e2e=7660fed
366	47.5752	127.0.0.1	127.0.0.1	DIAMETER	152	SACK cmd-Device-Watchdog Request(280) flags=R--- appl=Diameter Common Messages(0) h2h=104b0355 e2e=104b0355
381	47.5761	127.0.0.1	127.0.0.1	DIAMETER	168	SACK cmd-Device-Watchdog Answer(280) flags=---- appl=Diameter Common Messages(0) h2h=104b0355 e2e=104b0355
384	47.5765	127.0.0.1	127.0.0.1	DIAMETER	164	SACK cmd-Device-Watchdog Answer(280) flags=---- appl=Diameter Common Messages(0) h2h=7660fed e2e=7660fed
679	77.5804	127.0.0.1	127.0.0.1	DIAMETER	140	cmd-Device-Watchdog Request(280) flags=R--- appl=Diameter Common Messages(0) h2h=7660fee e2e=7660fee
688	77.5818	127.0.0.1	127.0.0.1	DIAMETER	164	SACK cmd-Device-Watchdog Answer(280) flags=---- appl=Diameter Common Messages(0) h2h=7660fee e2e=7660fee
736	81.0369	127.0.0.1	127.0.0.1	DIAMETER	764	cmd-Re-Auth Request(258) flags=RP-- appl=3GPP Gx(16777238) h2h=104b0356 e2e=104b0356
745	81.0440	127.0.0.1	127.0.0.1	DIAMETER	496	SACK cmd-Re-Auth Answer(258) flags=-P- appl=3GPP Gx(16777238) h2h=104b0356 e2e=104b0356
931	107.598	127.0.0.1	127.0.0.1	DIAMETER	140	cmd-Device-Watchdog Request(280) flags=R--- appl=Diameter Common Messages(0) h2h=7660fef e2e=7660fef
940	107.599	127.0.0.1	127.0.0.1	DIAMETER	164	SACK cmd-Device-Watchdog Answer(280) flags=---- appl=Diameter Common Messages(0) h2h=7660fef e2e=7660fef
11	137.631	127.0.0.1	127.0.0.1	DIAMETER	136	cmd-Device-Watchdog Request(280) flags=R--- appl=Diameter Common Messages(0) h2h=104b0357 e2e=104b0357
11	137.632	127.0.0.1	127.0.0.1	DIAMETER	156	SACK cmd-Device-Watchdog Request(280) flags=R--- appl=Diameter Common Messages(0) h2h=7660ff0 e2e=7660ff0
12	137.636	127.0.0.1	127.0.0.1	DIAMETER	164	SACK cmd-Device-Watchdog Answer(280) flags=---- appl=Diameter Common Messages(0) h2h=7660ff0 e2e=7660ff0
12	137.636	127.0.0.1	127.0.0.1	DIAMETER	168	SACK cmd-Device-Watchdog Answer(280) flags=---- appl=Diameter Common Messages(0) h2h=104b0357 e2e=104b0357

Frame 679: 140 bytes on wire (1120 bits), 140 bytes captured (1120 bits) on interface 0

- Linux cooked capture
- Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
- Stream Control Transmission Protocol, Src Port: 3868 (3868), Dst Port: 35002 (35002)
- Diameter Protocol
 - Version: 0x01
 - Length: 76
 - Flags: 0x80, Request
 - Command Code: 280 Device-Watchdog
 - ApplicationId: Diameter Common Messages (0)
 - Hop-by-Hop Identifier: 0x07660fee
 - End-to-End Identifier: 0x07660fee
 - [Answer In: 688]
 - AVP: Origin-Host(264) l=22 f=M- val=pgw.le.nsn.com
 - AVP: Origin-Realm(296) l=18 f=M- val=le.nsn.com
 - AVP: Origin-State-Id(278) l=12 f=M- val=1539264682



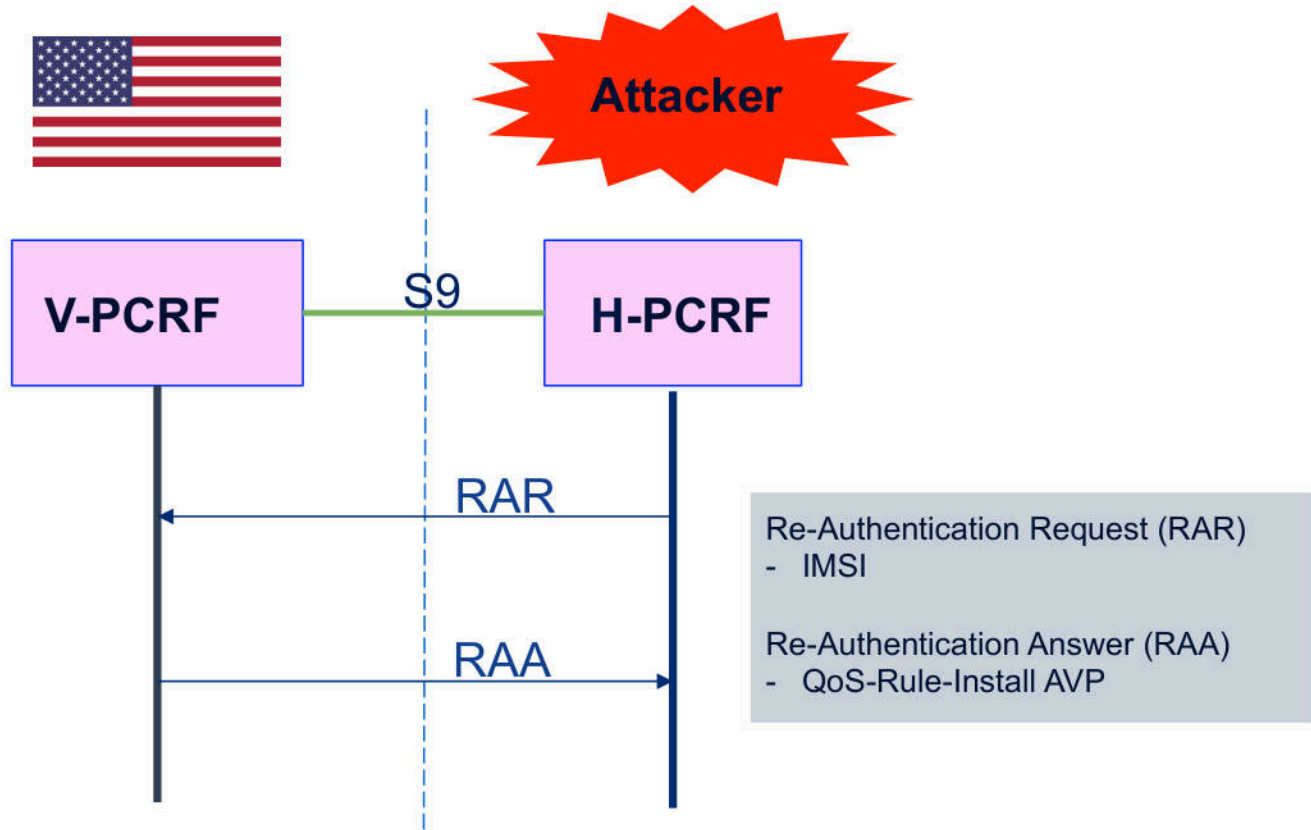
Attack Scenario 1: Putting PCC via RAR (posing as home network)



Re-Authentication Request
- QoS-Rule-Install

Answer does not matter

Attack Scenario 2: Putting PCC via RAR to outgoing roamer



UEPROC [1] ./ueproc 1

```
29.6.2018 11:36:05,015,233 UE-C-1 NAS PROCEDURE COMPLETED ATTACH IMSI=588711002000111
29.6.2018 11:36:05,016,253 UE-C-1 RRC ENB-C-1 MESSAGE SENT RRC_UL_INFORMATION_TRANSFER
29.6.2018 11:36:05,017,142 UE-C-1 RRC ENB-C-1 MESSAGE RECEIVED RRC_UE_CAPABILITY_ENQUIRY
29.6.2018 11:36:05,017,258 UE-C-1 RRC ENB-C-1 MESSAGE SENT RRC_UE_CAPABILITY_INFORMATION
29.6.2018 11:36:05,265,910 UE-C-1 RRC ENB-C-1 MESSAGE RECEIVED RRC_DL_INFORMATION_TRANSFER
```

UEUPROC [1] ./ueuproc 1

```
: set echo off
29.6.2018 11:21:25,211,737 UE-U-1 - PROCESS STARTED
Waiting for inputs ...
29.6.2018 11:33:20,722,736 UE-U-1 SAI PROCEDURE COMPLETED L1_CELL_SEARCH "ENB-ID = 1"
29.6.2018 11:36:03,906,972 UE-U-1 SAI PROCEDURE COMPLETED L1_CELL_SEARCH "ENB-ID = 1"
```

SGW [1] ./sgw 1

```
29.6.2018 11:36:05,261,967 SGW-1 GTP_S11 PROCEDURE STARTED MODIFY_BEARER IMSI=588711002000111 R-TEID=8439 S-TEID=8434
29.6.2018 11:36:05,262,010 SGW-1 GTP_S11 PROCEDURE COMPLETED MODIFY_BEARER IMSI=588711002000111 R-TEID=8434 S-TEID=8439
29.6.2018 11:36:05,262,069 SGW-1 GTP_S11 MME-1 MESSAGE SENT GTPV2_PDU_MODIFY_BEARER_RESPONSE
```

ERIM [1] ./erim 1

```
: set echo off
29.6.2018 11:21:20,276,273 ERIM-1 - PROCESS STARTED

Running in SAI mode
Waiting for inputs ...
29.6.2018 11:21:24,932,897 ERIM-1 SAI ENB-U-1 MESSAGE RECEIVED ENB_REGISTER
```

PGW [1] ./pgw 1

```
29.6.2018 13:15:54,464,586 PGW-1 DIAMETER PCRF-1 MESSAGE RECEIVED DIAMETER_DWA
29.6.2018 13:16:20,470,259 PGW-1 DIAMETER_Gx PCRF-1 MESSAGE RECEIVED DIAMETER_Gx_RAR IMSI=588711002000111
29.6.2018 13:16:20,470,424 PGW-1 DIAMETER_Gx PROCEDURE STARTED RE_AUTH IMSI=588711002000111
29.6.2018 13:16:20,477,998 PGW-1 DIAMETER_Gx PROCEDURE COMPLETED RE_AUTH IMSI=588711002000111
29.6.2018 13:16:20,478,086 PGW-1 DIAMETER_Gx PCRF-1 MESSAGE SENT DIAMETER_Gx_RAH IMSI=588711002000111
29.6.2018 13:16:24,469,085 PGW-1 DIAMETER PCRF-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 13:16:24,469,513 PGW-1 DIAMETER PCRF-1 MESSAGE SENT DIAMETER_DWA
```

ENBC [1] ./enbc 1

```
_INFORMATION
29.6.2018 11:36:05,057,160 ENB-C-1 S1-AP PROCEDURE COMPLETED INITIAL_CONTEXT_SETUP
29.6.2018 11:36:05,057,368 ENB-C-1 S1-AP MME-1 MESSAGE SENT INITIAL_CONTEXT_SETUP_RESPONSE
29.6.2018 11:36:05,057,886 ENB-C-1 S1-AP MME-1 MESSAGE SENT UPLINK_NAS_TRANSPORT
29.6.2018 11:36:05,263,828 ENB-C-1 S1-AP MME-1 MESSAGE RECEIVED DOWNLINK_NAS_TRANSPORT
29.6.2018 11:36:05,264,621 ENB-C-1 RRC UE-C-1 MESSAGE SENT RRC_DL_INFORMATION_TRANSFER
```

ENBU [1] ./enbu 1

```
29.6.2018 11:21:23,973,567 ENB-U-1 - PROCESS STARTED
Waiting for inputs ...
29.6.2018 11:21:24,932,414 ENB-U-1 SAI ERIM-1 MESSAGE SENT ENB_REGISTER
29.6.2018 11:33:20,722,856 ENB-U-1 SAI PROCEDURE COMPLETED L1_CELL_SEARCH "UE-ID = 1"
29.6.2018 11:36:03,907,125 ENB-U-1 SAI PROCEDURE COMPLETED L1_CELL_SEARCH "UE-ID = 1"
```

MME [1] ./mme 1

```
29.6.2018 13:15:20,799,137 MME-1 DIAMETER HSS-1 MESSAGE SENT DIAMETER_DWR
29.6.2018 13:15:20,801,669 MME-1 DIAMETER HSS-1 MESSAGE RECEIVED DIAMETER_DWA
29.6.2018 13:15:50,807,623 MME-1 DIAMETER HSS-1 MESSAGE SENT DIAMETER_DWR
29.6.2018 13:15:50,810,123 MME-1 DIAMETER HSS-1 MESSAGE RECEIVED DIAMETER_DWA
29.6.2018 13:16:20,814,006 MME-1 DIAMETER HSS-1 MESSAGE SENT DIAMETER_DWR
29.6.2018 13:16:20,816,962 MME-1 DIAMETER HSS-1 MESSAGE RECEIVED DIAMETER_DWA
```

HSS [1] ./hss 1

```
29.6.2018 13:14:50,796,827 HSS-1 DIAMETER MME-1 MESSAGE SENT DIAMETER_DWA
29.6.2018 13:15:20,800,340 HSS-1 DIAMETER MME-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 13:15:20,800,576 HSS-1 DIAMETER MME-1 MESSAGE SENT DIAMETER_DWA
29.6.2018 13:15:50,808,785 HSS-1 DIAMETER MME-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 13:15:50,809,346 HSS-1 DIAMETER MME-1 MESSAGE SENT DIAMETER_DWA
29.6.2018 13:16:20,815,390 HSS-1 DIAMETER MME-1 MESSAGE RECEIVED DIAMETER_DWR
29.6.2018 13:16:20,815,695 HSS-1 DIAMETER MME-1 MESSAGE SENT DIAMETER_DWA
```

PCRF [1] ./pcrf 1

```
: 29.6.2018 13:16:20,467,952 PCRF-1 DIAMETER_Gx PROCEDURE STARTED RE_AUTH IMSI=588711002000111
29.6.2018 13:16:20,468,170 PCRF-1 DIAMETER_Gx PGW-1 MESSAGE SENT DIAMETER_Gx_RAR IMSI=588711002000111
Waiting for inputs ...
29.6.2018 13:16:20,479,368 PCRF-1 DIAMETER_Gx PGW-1 MESSAGE RECEIVED DIAMETER_Gx_RAH IMSI=588711002000111
29.6.2018 13:16:20,479,402 PCRF-1 DIAMETER_Gx PROCEDURE COMPLETED RE_AUTH IMSI=588711002000111
29.6.2018 13:16:24,467,829 PCRF-1 DIAMETER PGW-1 MESSAGE SENT DIAMETER_DWR
29.6.2018 13:16:24,471,045 PCRF-1 DIAMETER PGW-1 MESSAGE RECEIVED DIAMETER_DWA
```

diameter

No.	Time	Source	Destination	Protocol	Length	Info
39..	2749.22..	127.0.0.1	127.0.0.1	DIAMETER	140	cmd-Device-Watchdog Request(280) flags=R--- appl=Diameter Common Messages(0) h2h=7661030 e2e=7661030
39..	2749.22..	127.0.0.1	127.0.0.1	DIAMETER	152	SACK cmd-Device-Watchdog Request(280) flags=R--- appl=Diameter Common Messages(0) h2h=104b0388 e2e=104b0388
39..	2749.22..	127.0.0.1	127.0.0.1	DIAMETER	168	SACK cmd-Device-Watchdog Answer(280) flags=---- appl=Diameter Common Messages(0) h2h=104b0388 e2e=104b0388
39..	2749.22..	127.0.0.1	127.0.0.1	DIAMETER	164	SACK cmd-Device-Watchdog Answer(280) flags=---- appl=Diameter Common Messages(0) h2h=7661030 e2e=7661030
40..	2775.23..	127.0.0.1	127.0.0.1	DIAMETER	704	cmd-Re-Auth Request(258) flags=RP-- appl=3GPP Gx(16777238) h2h=104b0389 e2e=104b0389
40..	2775.24..	127.0.0.1	127.0.0.1	DIAMETER	496	SACK cmd-Re-Auth Answer(258) flags=-P- appl=3GPP Gx(16777238) h2h=104b0389 e2e=104b0389
40..	2779.23..	127.0.0.1	127.0.0.1	DIAMETER	168	SACK cmd-Device-Watchdog Request(280) flags=---- appl=Diameter Common Messages(0) h2h=104b038a e2e=104b038a
40..	2809.26..	127.0.0.1	127.0.0.1	DIAMETER	140	cmd-Device-Watchdog Request(280) flags=R--- appl=Diameter Common Messages(0) h2h=7661031 e2e=7661031
40..	2809.26..	127.0.0.1	127.0.0.1	DIAMETER	164	SACK cmd-Device-Watchdog Answer(280) flags=---- appl=Diameter Common Messages(0) h2h=7661031 e2e=7661031
40..	2810.48..	127.0.0.1	127.0.0.1	DIAMETER	704	cmd-Re-Auth Request(258) flags=RP-- appl=3GPP Gx(16777238) h2h=104b038b e2e=104b038b
40..	2810.49..	127.0.0.1	127.0.0.1	DIAMETER	496	SACK cmd-Re-Auth Answer(258) flags=-P- appl=3GPP Gx(16777238) h2h=104b038b e2e=104b038b
40..	2839.26..	127.0.0.1	127.0.0.1	DIAMETER	140	cmd-Device-Watchdog Request(280) flags=R--- appl=Diameter Common Messages(0) h2h=7661032 e2e=7661032
40..	2839.26..	127.0.0.1	127.0.0.1	DIAMETER	164	SACK cmd-Device-Watchdog Answer(280) flags=---- appl=Diameter Common Messages(0) h2h=7661032 e2e=7661032

▶ Frame 40216: 136 bytes on wire (1088 bits), 136 bytes captured (1088 bits) on interface 0
 ▶ Linux cooked capture
 ▶ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
 ▶ Stream Control Transmission Protocol, Src Port: 35902 (35902), Dst Port: 3868 (3868)
 ▶ Diameter Protocol
 Version: 0x01
 Length: 72
 Flags: 0x80, Request
 Command Code: 280 Device-Watchdog
 ApplicationId: Diameter Common Messages (0)
 Hop-by-Hop Identifier: 0x104b038a
 End-to-End Identifier: 0x104b038a
 [Answer In: 40225]
 ▶ AVP: Origin-Host(264) l=23 f=M- val=pcrf1.le.nsn.com
 ▶ AVP: Origin-Realm(296) l=16 f=M- val=pcrf1.le
 ▶ AVP: Origin-State-Id(278) l=12 f=M- val=1539267384

Before and After

.RaaChargingRuleBaseName1_1	rulebasename-11
.RaaChargingRuleBaseName1_2	rulebasename-12
.RaaChargingRuleBaseName2_1	rulebasename-21
.RaaChargingRuleBaseName2_2	rulebasename-22
.RaaChargingRuleName1_1	\7063632d31
.RaaChargingRuleName1_2	\7063632d32 ←
.RaaChargingRuleName2_1	\7063632d33
.RaaChargingRuleName2_2	\7063632d34

.RaaChargingRuleBaseName1_1	rulebasename-11
.RaaChargingRuleBaseName1_2	rulebasename-12
.RaaChargingRuleBaseName2_1	rulebasename-21
.RaaChargingRuleBaseName2_2	rulebasename-22
.RaaChargingRuleName1_1	\7063632d31
.RaaChargingRuleName1_2	\7063632d31 ←
.RaaChargingRuleName2_1	\7063632d31
.RaaChargingRuleName2_2	\7063632d31

Impacts

- Attacker:
 - Better services
 - Shifting the costs – Letting somebody else pay the phone bill
 - Re-selling "opportunity"
- Users:
 - Might be billed for services he has not used (in particular company subscriptions are at risk)
- Operators:
 - Bill disputes (service desks)
 - Loss of corporate customers
 - Costs with partners that can not be charged to a user
 - IPX carriers still want to see their money

Countermeasures

Switch it off – build it from scratch?



Countermeasures

For Operators

- S9 Interface -> use IPSec with trusted partners directly
- S9 only open on need basis
- Routing via origin realm, origin host
- IMSI range – operator match
- Check not to get messages from yourself
- Logical separation of visitors and own subscribers
- Location distance
- Fingerprint partner
- Fingerprint "flows"

For "normal" Users

- Check your bill
- Keep an eye on the news

For "corporate" Users

- Security and network protection is something that needs to be part of a Service Layer Agreement
- It is a quality indicator, similar to bandwidth and coverage

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Questions?

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